



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

MATHEMATICS

9709/01

Paper 1 Pure Mathematics 1 (P1)

October/November 2007

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

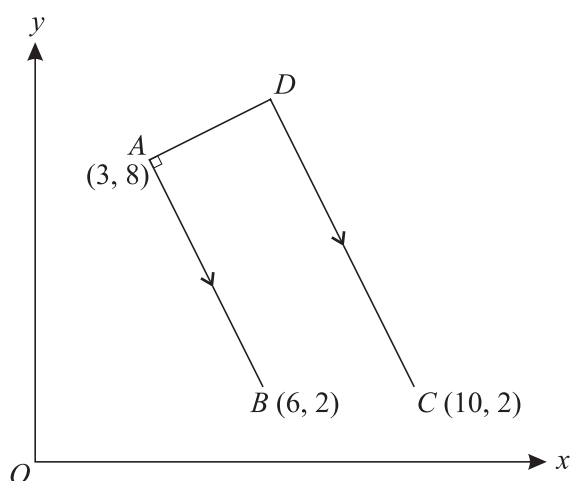
Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.



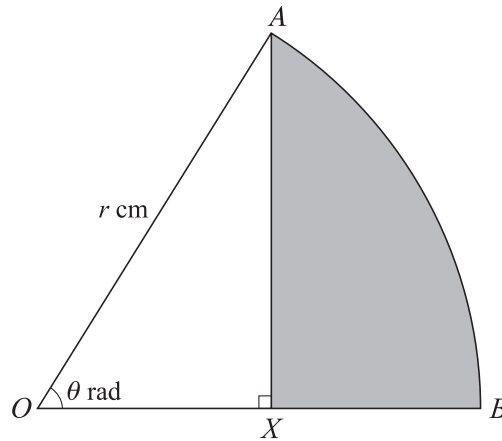
- 1 Determine the set of values of the constant k for which the line $y = 4x + k$ does not intersect the curve $y = x^2$. [3]
- 2 Find the area of the region enclosed by the curve $y = 2\sqrt{x}$, the x -axis and the lines $x = 1$ and $x = 4$. [4]
- 3 (i) Find the first three terms in the expansion of $(2 + u)^5$ in ascending powers of u . [3]
- (ii) Use the substitution $u = x + x^2$ in your answer to part (i) to find the coefficient of x^2 in the expansion of $(2 + x + x^2)^5$. [2]
- 4 The 1st term of an arithmetic progression is a and the common difference is d , where $d \neq 0$.
- (i) Write down expressions, in terms of a and d , for the 5th term and the 15th term. [1]
- The 1st term, the 5th term and the 15th term of the arithmetic progression are the first three terms of a geometric progression.
- (ii) Show that $3a = 8d$. [3]
- (iii) Find the common ratio of the geometric progression. [2]
- 5 (i) Show that the equation $3 \sin x \tan x = 8$ can be written as $3 \cos^2 x + 8 \cos x - 3 = 0$. [3]
- (ii) Hence solve the equation $3 \sin x \tan x = 8$ for $0^\circ \leq x \leq 360^\circ$. [3]

6



The three points $A(3, 8)$, $B(6, 2)$ and $C(10, 2)$ are shown in the diagram. The point D is such that the line DA is perpendicular to AB and DC is parallel to AB . Calculate the coordinates of D . [7]

7



In the diagram, AB is an arc of a circle, centre O and radius r cm, and angle $AOB = \theta$ radians. The point X lies on OB and AX is perpendicular to OB .

- (i) Show that the area, $A \text{ cm}^2$, of the shaded region AXB is given by

$$A = \frac{1}{2}r^2(\theta - \sin \theta \cos \theta). \quad [3]$$

- (ii) In the case where $r = 12$ and $\theta = \frac{1}{6}\pi$, find the perimeter of the shaded region AXB , leaving your answer in terms of $\sqrt{3}$ and π . [4]

- 8 The equation of a curve is $y = (2x - 3)^3 - 6x$.

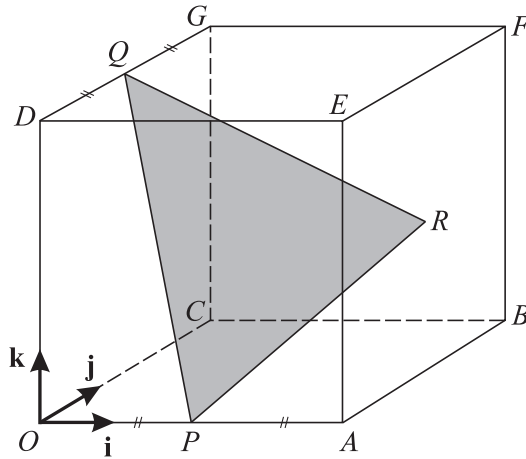
- (i) Express $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of x . [3]

- (ii) Find the x -coordinates of the two stationary points and determine the nature of each stationary point. [5]

- 9 A curve is such that $\frac{dy}{dx} = 4 - x$ and the point $P(2, 9)$ lies on the curve. The normal to the curve at P meets the curve again at Q . Find

- (i) the equation of the curve, [3]
 (ii) the equation of the normal to the curve at P , [3]
 (iii) the coordinates of Q . [3]

10



The diagram shows a cube $OABCDEFG$ in which the length of each side is 4 units. The unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to $\overrightarrow{OA}$, $\overrightarrow{OC}$ and $\overrightarrow{OD}$ respectively. The mid-points of OA and DG are P and Q respectively and R is the centre of the square face $ABFE$.

(i) Express each of the vectors $\overrightarrow{PR}$ and $\overrightarrow{PQ}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. [3]

(ii) Use a scalar product to find angle QPR . [4]

(iii) Find the perimeter of triangle PQR , giving your answer correct to 1 decimal place. [3]

11 The function f is defined by $f : x \mapsto 2x^2 - 8x + 11$ for $x \in \mathbb{R}$.

(i) Express $f(x)$ in the form $a(x + b)^2 + c$, where a , b and c are constants. [3]

(ii) State the range of f . [1]

(iii) Explain why f does not have an inverse. [1]

The function g is defined by $g : x \mapsto 2x^2 - 8x + 11$ for $x \leq A$, where A is a constant.

(iv) State the largest value of A for which g has an inverse. [1]

(v) When A has this value, obtain an expression, in terms of x , for $g^{-1}(x)$ and state the range of g^{-1} . [4]

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